

Multi-classification for EEG Motor Imagery Signals using Auto-selected Filter Bank Regularized Common Spatial Pattern

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Abstract— Motor Imagery (MI) is a critical topic in Brain-Computer Interface (BCI). Due to the low signal-to-noise ratio, it is not easy to accurately classify motor imagery signals, especially for multiple classification tasks. Common Spatial Pattern (CSP) is a spatial transformation method that can effectively extract spatial features of EEG signals. However, the covariance matrix is inaccurate due to the small training data size. Thus, in this paper, a regularization parameter auto-selection algorithm is proposed to automatically adjust the ratio of the covariance matrix calculated by other subjects' data based on the mutual information. It can be used to tackle the problem of an inaccurate mixed covariance matrix caused by fixed regularization parameters.

To illustrate the merits of the proposed Auto-selected Filter Bank Regularized Common Spatial Pattern (AFBRCSPP), we used the ten folds cross-validation accuracy and Kappa as the evaluation metrics to evaluate two data sets (BCI4-2a and BCI3a data set). Both data set include four mental classes. By using BCI4-2a data set, we found that the mean accuracy of AFBRSP is 77.31% and the Kappa is 0.6975, which is higher than Filter Bank Regularized Common Spatial Pattern (FBRCSPP) by 5.67% and 0.0756, respectively. By using BCI3a data set, the proposed AFBRCSPP improved the accuracy by 8.34% and the Kappa by 0.1111 compared with FBRCSPP where the mean accuracy of AFBRCSPP is 80.56%, and the kappa is 0.7407. The overall Kappa obtained by the proposed method is also higher than some state-of-the-art methods, implying that the proposed method is more reliable.

Keywords—Brain-Computer Interface, Motor imagery, Electroencephalogram

I. INTRODUCTION

Motor Imagery (MI) is one of the Brain-Computer Interface (BCI) applications which is the change of brain waves in the specific areas of the brain through imaginary movements without any actual actions. By acquiring and analyzing EEG signals, it can recognize the specific imagined actions. This technique can help disabled people do rehabilitation or control the wheelchair. Many devices can easily measure EEG signals. However, the signals are collected on the skin of the brain and cause EEG signals to be feeble and easily affected by various noises. This also makes it difficult to analyze and identify the

EEG signal. Especially for motor imagery, different people have different imagination abilities. Thus, the classification performance of motor imagery signals is not good.

EEG signal analysis methods commonly include time-domain analysis, frequency-domain analysis, time-frequency analysis, and spatial-domain analysis [1]. One of the popular spatial analysis methods is Common Spatial Pattern (CSP). It maximizes the variance difference of the two types of EEG signals. [2] used the traditional CSP method to extract spatial features. They proposed a binary harmony search algorithm to select adequate channel data from spatial features and then used the extracted effective data to re-do CSP for spatial transformation to obtain more effective classification features. [3] mainly focused on reducing the dimension of EEG data. They first calculated the correlation between channel data and selected adequate channel data. After that, the covariance matrix feature was calculated. Finally, the EEG signal was spatially transformed by CSP.

Some researchers tend to combine CSP with some other methods, which can help to make up for the disadvantages of the CSP method. [4] integrated the CSP methods and local activities estimation. CSP is less sensitive to the location of the electrodes but sensitive to the size of the training set. Thus, they took advantage of both methods. They tested the proposed method using four public datasets. They demonstrated that combining two methods works better than using the two methods separately. [5] performed frequency domain filtering on EEG signals before using CSP. They designed a dual-tree complex wavelet transform-based filter bank, improving the spatial feature extraction efficiency. Meanwhile, they used neighbourhood component analysis to select the classification features.

Some papers improved the CSP algorithm. [6] introduced a sliding time window to divide the data into multiple periods before using CSP. They proposed two sliding window methods. After using the time window, more efficient features can be obtained by using CSP. [7] proposed a correlation-based CSP. When constructing the CSP filter matrix, they calculated the similarity between the signals and used it as a penalty term to adjust the CSP filter matrix. They tested three public datasets and compared the proposed method to other CSP-related methods. From the results, correlation-based CSP can improve classification accuracy. [8] simultaneously filtered the EEG

signal in time, frequency, and spatial domain. First, they used multiple frequency filters to divide the EEG signal into sub-band signals. Then, they introduced sliding time windows to subdivide the signal in time for each frequency band signal. Finally, CSP was used for each time segment of each frequency segment to obtain multiple sets of spatial features. The effective spatial features were selected as the final classification features.

Although CSP is an effective method to extract EEG features, this algorithm has some drawbacks. CSP method is a supervised learning method. It requires lots of training data. If the training sample is not adequate, the obtained spatial filter may not accurate. In practical applications, long-time data collection may make subjects feel tired. Thus, whether it is a public data set or self-collected data, the amount of data for each motor imagery is not large. Using small sample data may lead to an inaccurate covariance matrix. Regularized Common Spatial Pattern (RCSP) can solve this problem. The difference between RCSP and CSP is that RCSP adds a certain amount of other subject samples and adds a penalty term to correct the covariance matrix. However, when using RCSP, adding too many samples from other subjects may lead to the loss of information in the target data. Similarly, the scale of the penalty term affects the accuracy of the covariance. Thus, how to control the proportion of adding other samples and the proportion of the correction penalty item is a complex problem.

In this paper, we propose an Auto-selected Filter Bank Regularized Common Spatial Pattern (AFBRCSP) algorithm, which can automatically select the regularization parameters. Two regularization parameters are used to control the proportion of additional data from other subjects and correct the error of the mixed covariance matrix. We use mutual information to evaluate the degree of the difference among the generated features by using multiple groups of regularization parameters. The mutual information estimation matrix can be used to determine whether the selected regularization parameters are appropriate to the mixed covariance matrix. The distribution of the filtered data selected by this method is close to the distribution of the target classification features. Thus, the spatial features obtained by the selected regularization parameters are more suitable for the network to do classification. In addition, it can automatically adjust the parameters based on the extracted features from different subjects for different motor imagery tasks. This method significantly improves the accuracy compared to traditional FBCSP using the fixed regularization parameters.

II. METHODOLOGY

A. Overview of system

The block diagram of the entire system is presented in Fig. 1. We divide the target subject's data into the training set and testing set and prepare other subjects' data. All data are pre-processed, including re-reference and average removal. Firstly, multiple band-pass filters are used to filter the training data and testing data. Secondly, training data and other subjects' EEG data are used to initialize the spatial filter matrix. Next, the spatially filtered features are obtained and mutual information is used to evaluate these features and select suitable regularization parameters. The parameters are employed to rebuild the RCSP filters. When the testing data come, the new RCSP filters are

applied to the data. Finally, the dimension of the RCSP features of the testing data has been reduced and he selected multiple groups of features to feed into the CNN to classify the motor imagery tasks.

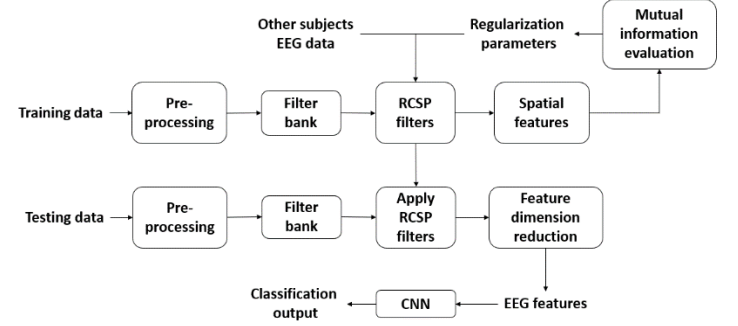


Fig. 1. Block diagram of the entire system

B. Regularized Common Spatial Pattern

Regularized common Spatial Pattern (RCSP) is a method that adds other subjects' data to extract effective components to maximize the variance difference of two signals. Consider S_c is the original EEG data and $\hat{S}_c$ is the EEG data of other subjects, we have:

$$R_c = \sum_{n=1}^{N_t} \frac{S_{cn} S_{cn}^T}{\text{trace}(S_{cn} S_{cn}^T)} \quad (1)$$

$$\hat{R}_c = \sum_{\hat{n}=1}^{\hat{N}_t} \frac{\hat{S}_{c\hat{n}} \hat{S}_{c\hat{n}}^T}{\text{trace}(\hat{S}_{c\hat{n}} \hat{S}_{c\hat{n}}^T)} \quad (2)$$

where R_c is the covariance of S_c and $\hat{R}_c$ is the covariance of $\hat{S}_c$; $\text{trace}(S_c)$ is the sum of elements on the diagonal of the matrix S_c ; N_t is the number of trials of S_c ; $\hat{N}_t$ is the number of trials of $\hat{S}_c$. We have:

$$J_c(\beta_c) = \frac{(1-\beta_c) \cdot R_c + \beta_c \cdot \hat{R}_c}{(1-\beta_c) \cdot N_t + \beta_c \cdot \hat{N}_t} \quad (3)$$

$$\Sigma_c(\beta_c, \gamma_c) = (1 - \gamma_c) \cdot J_c(\beta_c) + \frac{\gamma_c}{N_c} \text{trace}[J_c(\beta_c)] \cdot I \quad (4)$$

where N_c is the total channel numbers; β_c controls the variance of the estimated covariance; γ_c is the second regularized parameter, which can reduce large eigenvalues and increase small eigenvalues; $\Sigma(\beta_c, \gamma_c)$ is the mixed covariance matrix. Then, decompose mixed covariance matrix. It gets eigenvalue λ_c and eigenvector U_c . Sort eigenvalue U_c in descending order and get the whitening matrix P_w .

$$\Sigma_c = \Sigma_{c_1} + \Sigma_{c_2} = U_c \lambda_c U_c^T \quad (5)$$

$$P_w = \sqrt{\lambda_c^{-1}} U_c^T \quad (6)$$

where Σ_{c_1} is the mixed covariance matrix of the first-class data; Σ_{c_2} is the mixed covariance matrix of the second-class data. Apply P_w to the two classes mixed matrix, and then continue to decompose one of the class matrixes.

$$S_{w_1} = P_w \Sigma_{c_1} P_w^T \quad (7)$$

$$S_{w_2} = P_w \Sigma_{c_2} P_w^T \quad (8)$$

$$S_{w_1} = U_b \lambda_B U_b^T \quad (9)$$

where S_{w_1} is the whiten matrix of the first-class data; S_{w_2} is the whiten matrix of the second-class data. We can obtain the space filter, the filtered data matrix, and the feature matrix.

$$W_c = U_b^T P_w \quad (10)$$

$$Z_c = W_c * S_c \quad (11)$$

$$X_c = \text{var}(Z_c) \quad (12)$$

where Z_c is spatial filtered data; X_c is feature matrix; W_c is the RCSP filter; $\text{var}()$ is the function of calculating variance.

C. Auto-selected RCSP weights selection

After using the RCSP spatial filter, the feature vector is X_c , and the label vector is Y_c . Both vectors information entropy $H_I(X_c)$ and $H_I(Y_c)$ can be calculated. Then, use their joint probability density function to calculate their mutual information. For the probability density, firstly, the numerical range of the feature matrix is obtained, and then this range is segmented. Calculates the probability distribution based on the number of values that fall within each range, which is used as the probability density.

$$H_I(X_c) = -\sum_{x \in X_c} P(x) \log_2 P(x) \quad (13)$$

$$H_I(Y_c) = -\sum_{y \in Y_c} P(y) \log_2 P(y) \quad (14)$$

$$M_I(X_c, Y_c) = \frac{2 \sum_{y \in Y_c} \sum_{x \in X_c} P(x, y) \log \left(\frac{P(x, y)}{P(x)P(y)} \right)}{H_I(X_c) + H_I(Y_c)} \quad (15)$$

where $p(x)$ is the probability of x ; $p(y)$ is the probability of y ; $p(x, y)$ is the joint probability of x and y . We assume that the X_c and Y_c are relatively independent, then $p(x, y) = p(x)p(y)$. The final calculated mutual information $M_I(X_c, Y_c) = 0$. If the value of M_I is larger, the feature is closely related to the label. In this way, M_I can be used to evaluate whether the obtained EEG signal characteristics are suitable for distinguishing the two categories. The two parameters of RCSP can take values within a certain range. Then calculate the mutual information matrix of the feature labels of each set of RCSP parameters. In this matrix, the γ_c and β_c corresponding to the maximum value are used as the final RCSP parameters.

III. EXPERIMENTS AND RESULTS

A. Data description

The first data set is BCI Competition 4-2a (BCI4-2a) [9]. This data set has nine subjects, 22 channels, and 4 imaginary movement commands, which are left-hand, right-hand, foot, and tongue imaginary movement. The second data set is BCI

Competition 3a (BCI3a) [10]. This data set has three subjects, 60 channels, and 4 imaginary movement commands, which are the same as the BCI competition 4-2a data set. However, they have different trial numbers. For BCI4-2a, there are 288 trial for each subject. For BCI3a, the first subject has 180 trials and the second and the third subject have 120 trials. The specific information of these two data sets is shown in Table I.

TABLE I. DESCRIPTION OF TWO PUBLIC DATA SETS

	Data set name	
	BCI4-2a	BCI3a
Trial number (total)	288 (each subject)	180/120/120
Trial number (each)	72	45/30/30
Class number	4	4
Subject number	9	3

B. Evaluation methods

In the experiments, we use ten cross-validation accuracy as the evaluation method. In addition, we also use the Cohen's kappa coefficient to evaluate the performance of the proposed method. Kappa is a statistic that is used to measure inter-rater reliability for categorical items [11]. The calculating method of Kappa κ and standard error of Kappa se is shown in (17) and (18).

$$p_e = \frac{\sum_i n_{ki} * n_{ik}}{N_g * N_g} \quad (16)$$

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (17)$$

$$se(\kappa) = \frac{\sqrt{p_0 - p_e^2 - \sum_i [n_{ki} * n_{ik} * (n_{ki} + n_{ik})] / N_p^3}}{(1 - p_e) \sqrt{N_p}} \quad (18)$$

where n_{ki} is the number of items classified as k by label i , while n_{ik} is the number of items classified as i by label k ; N_g is the total items to classify; p_e is a probability of agreement by chance; p_0 is the relative observed agreement; N_p is the total number of samples.

C. Experiments

1) Experiment A: Regularized FBCSP

First, we use regularized FBCSP as the feature extraction method and use CNN as the classifier. We first filter the EEG signal into ten frequency bands, namely 1 Hz to 4 Hz, 4 Hz to 8 Hz, 8 Hz to 12 Hz, 12 Hz to 16 Hz, 16 Hz to 20 Hz, 20 Hz to 24 Hz, 24 Hz to 28 Hz, 28 Hz to 32 Hz, 32 Hz to 36 Hz, 36 Hz to 40 Hz. Then, for each band, we apply RCSP to each sub-signal. In this experiment, γ_c is set to 0.1, β_c is set to 0.2.

2) Experiment B: Auto-selected Regularized FBCSP

We introduce a method that can automatically adjust the two parameters in this experiment. First, filter the data in defined frequency bands. Before building the CSP matrix, calculate other subjects' data covariance and use equation (4) to get the

mixed covariance matrix. In this step, we use multiple groups of β_c and γ_c , the range of the two parameters is between 0 and 1. γ_c takes [0,0.1,0.2,0.3] and β_c takes [0,0.1,0.2,0.3,0.4].

Then, we use β_c and γ_c to calculate the CSP spatial filters. There is a total of 20 sets of spatial filters which are obtained in each frequency band. The 20 sets of filters are applied to the original data to obtain the spatially filtered data. The variance of the filtered data is calculated as the features. Use the variance features to calculate the mutual information following the equation (15). The larger mutual information means the β_c and γ_c of this set of RCSP filters is better. Therefore, use this group of β_c and γ_c as the final regularization parameters. The results of two experiments are shown in Table II and Table III.

TABLE II. PERFORMANCE FOR BCI4-2A USING REGULARIZED FBCSP AND AUTO-SELECTED REGULARIZED FBCSP WITH CNN

	FBCSP			Auto-selected FBCSP		
	Accuracy (%)	Kappa	Se	Accuracy (%)	Kappa	Se
Sub 1	85.07	0.8009	0.0640	87.85	0.8380	0.0653
Sub 2	64.24	0.5231	0.0528	71.88	0.6250	0.0572
Sub 3	88.89	0.8519	0.0658	93.75	0.9167	0.0680
Sub 4	56.60	0.4213	0.0483	58.68	0.4491	0.0496
Sub 5	47.92	0.3056	0.0424	61.11	0.4815	0.0511
Sub 6	48.26	0.3102	0.0427	52.08	0.3611	0.0453
Sub 7	88.89	0.8519	0.0658	93.75	0.9167	0.0680
Sub 8	81.94	0.7593	0.0625	89.24	0.8565	0.0660
Sub 9	82.99	0.7731	0.0630	87.50	0.8333	0.0651
Mean	71.64	0.6219	0.0564	77.31	0.6975	0.0595

TABLE III. PERFORMANCE FOR BCI3A USING REGULARIZED FBCSP AND AUTO-SELECTED REGULARIZED FBCSP WITH CNN

	FBCSP			Auto-selected FBCSP		
	Accuracy (%)	Kappa	Se	Accuracy (%)	Kappa	Se
K3b	88.33	0.8444	0.0829	93.33	0.9111	0.0858
K6b	58.33	0.4444	0.0765	65.83	0.5444	0.0835
L1b	70.00	0.6000	0.0871	82.50	0.7667	0.0971
Mean	72.22	0.6296	0.0822	80.56	0.7407	0.0888

IV. DISCUSSION

Regularization uses the other subjects' data and target data to jointly calculate the covariance matrix. It is possible that the selection of the regularization parameters β_c and γ_c are not suitable for the target subject's data. β_c is used to control the proportion of other subjects' data which are added to the calculation of the mixed covariance matrix. If β_c is large which implies that the model uses more other subjects' samples to calculate the mixed covariance matrix. However, the states and imagination abilities of different subjects are different.

Therefore, if we only use other subjects' data to calculate the covariance matrix, the spatial information distribution of other subjects may be different from the target subject, which may cause the generated spatial filter may not be suitable for the target subject's data. Eventually, it could lead to a decrease in classification accuracy.

In addition, γ_c is used to control the covariance error caused by the small value of the covariance matrix. This parameter does not provide much information on the original data covariance or newly added data covariance. It is only used to correct the deviation when adding other subjects' data. Therefore, if this parameter is large, the model correction is strong. The correction information may be increased, but lots of information of the original covariance matrix could be lost. Therefore, the selection of these two parameters largely affects the performance of the final spatial filter and indirectly affects the accuracy of the extracted features.

We propose a parameter auto-selected method to resolve this issue. Consider that if we can know whether each spatial filter generated using the selected group of regularization parameters can be highly distinguished before extracting the features, then we can use this group of regularization parameters to generate the best spatial filter. Mutual information can be used to evaluate the degree of the difference between the two groups of data. Thus, we can use mutual information to evaluate whether two groups of generated features can be better distinguished. If they can be easily distinguished, the mutual information should be large. In addition, we use the convolutional neural network as a classifier. The loss function is cross-entropy, which also belongs to another type of mutual information. Thus, when we use mutual information to evaluate the difference of the features, the features obtained are also helpful for the network to do classification.

Therefore, we can calculate the mutual information of the features obtained by using each regularisation parameter group and comparing the difference. The feature mutual information generated by different parameters can be arranged into a matrix. We can draw the map of the matrix. The mutual information map of one of the subjects is generated by using this subject training data. There are four motor imagery tasks. For each task, we take β_c from [0, 0.4] and γ_c from [0, 0.3]. The maps are shown in Fig. 2.

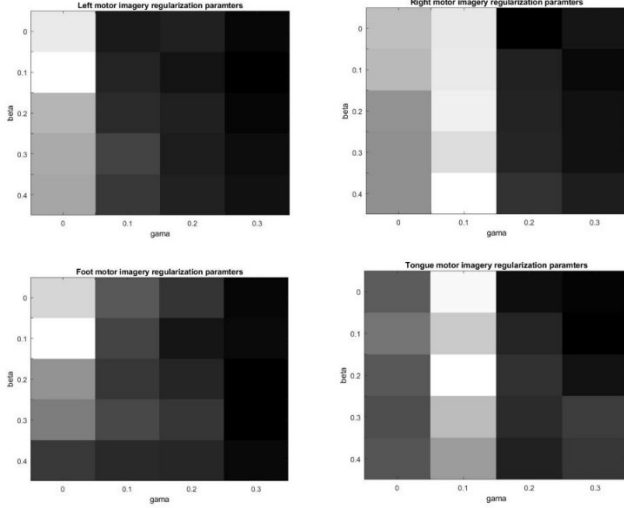


Fig. 2. Mutual information maps for different motor imagery tasks

In the map, the light color represents the significant mutual information while dark color represents the low mutual information. Large mutual information represents a higher degree of discrimination of the extracted features. From the map, the regularization parameter β_c of most of the better features produced is between $[0, 0.2]$, γ_c is between $[0, 0.1]$. However, for the right motor imagery task of this subject on the right-top in Fig. 2, the best β_c is 0.4 and γ_c is 0.1. This also proves that it should use different regularization parameters for different mental tasks. Thus, automatically selecting the regularization parameters according to the mutual information map can generate better spatial filters. It could more effectively apply the regularization function to the target data. From the results of experiments A and B (Table II, Table III), we can also see that the accuracy of using the AFBRCSP with auto-selected spatial filters is about 8% higher than that of the FBRCSF with the fixed parameters. It also proves that dynamically selecting the regularization parameter can generate more suitable spatial filters for the target data.

We also compare the results of our method with other methods proposed in recent years. We use kappa, and standard error as the evaluation measures. The comparison results are shown in Table IV, and Table V. From the results, we can see that the mean performance is better than others using both two data sets. Thus, these testing results support the original hypothesis that our proposed method is more effective to classify motor imagery multi-classification tasks.

TABLE IV. COMPARISON WITH OTHER PAPERS' METHODS FOR BCI4-2A (KAPPA)

BCI4-2a	Methods				
	<i>SS-MEMD BF [12]</i>	<i>TSSM+L DA [13]</i>	<i>3D CNN [14]</i>	<i>DPLM [15]</i>	<i>AFBRC SP+CN N</i>
Sub1	0.86	0.77	0.70	0.75	0.84
Sub2	0.24	0.33	0.46	0.49	0.63
Sub3	0.70	0.77	0.79	0.76	0.92
Sub4	0.68	0.51	0.59	0.49	0.45

BCI4-2a	Methods				
	<i>SS-MEMD BF [12]</i>	<i>TSSM+L DA [13]</i>	<i>3D CNN [14]</i>	<i>DPLM [15]</i>	<i>AFBRC SP+CN N</i>
Sub5	0.36	0.35	0.65	0.34	0.48
Sub6	0.34	0.36	0.54	0.36	0.36
Sub7	0.66	0.71	0.65	0.68	0.92
Sub8	0.75	0.72	0.70	0.76	0.86
Sub9	0.82	0.83	0.71	0.76	0.83
Mean $\pm$ std	0.60 $\pm$ 0.23	0.59 $\pm$ 0.21	0.64 $\pm$ 0.10	0.60 $\pm$ 0.18	0.70 $\pm$ 0.22

TABLE V. COMPARISON WITH OTHER PAPERS' METHODS FOR BCI3A (KAPPA)

BCI3a	Methods			
	<i>KLCSP [16]</i>	<i>LP-SVD +AR [17]</i>	<i>ARR [17]</i>	<i>AFBRCSP +CNN</i>
k3b	0.78	0.69	0.69	0.91
k6b	0.49	0.56	0.36	0.54
11b	0.56	0.45	0.39	0.77
Mean $\pm$ std	0.61 $\pm$ 0.15	0.57 $\pm$ 0.12	0.48 $\pm$ 0.18	0.74 $\pm$ 0.19

V. CONCLUSION

In this paper, we propose a regularization parameters auto-selection algorithm. This method solves the inaccurate covariance caused by the small sample data size. It can help select the appropriate proportion of newly added data and automatically select the appropriate proportion of covariance correction. We tested two public data sets to prove the feasibility of the proposed method. Compared to traditional CSP-related methods and classical motor imagery feature extraction methods proposed by other papers, our method can achieve higher performance. In the future, this method can be used on both offline and online motor imagery classification applications, especially when the system does not have adequate training data.

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